Draft version June 25, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

Learning the Effective PSF: A Neural PSF Field with Amortized Inference for DES Single-Epoch Imaging

Jeffrey Regier[1]
(The LSST Dark Energy Science Collaboration)

[1] *Department of Statistics, University of Michigan, Ann Arbor, MI 48109, USA*

ABSTRACT

Errors in the modeled point-spread-function (PSF) are a leading source of systematic error in weak-lensing shear measurement. Per-exposure fitters (e.g., PSFEx, PIFF) represent the PSF as a sub-pixel surface-brightness profile that must be convolved with an assumed pixel response to predict an image. We instead target the *effective* PSF (the pixel-convolved point-source response): the quantity sampled data identify and the exact kernel for forward-modeling galaxies. A single neural network (multi-head attention over an exposure's stars) predicts this effective PSF at any queried position and oversampling in one forward pass, *amortizing* the per-exposure fitting that PSFEx and PIFF repeat on every image. On 3,301 frozen test exposures of Dark Energy Survey single-epoch imaging, the model matches or exceeds PIFF and PSFEx on every reserved-star metric—size and ellipticity residuals, shape correlation, and per-pixel goodness of fit. In a star-anchored galaxy injection–recovery test on real data, it recovers galaxy *ellipticity*—the weak-lensing-critical quantity—without coherent bias and on par with PIFF, while recovered *sizes* run a few percent small. Higher-capacity coordinate encodings reduce this size deficit, but controlled simulations with known PSFs do not reproduce it at any decoder capacity, so it is specific to real data and its origin remains under investigation. Color conditioning further removes a chromatic PSF-size systematic that the baselines retain, and the model stays accurate in star-sparse exposures where PIFF degrades sharply. Controlled simulations motivate two training choices: a *blend forward-model* loss that removes a size bias from unrecognized blends, and point-source-only attention, which fixes a spin-2 representation failure triggered by galaxy detections in the context.

Keywords: Weak gravitational lensing (1797) — Point spread function (1349) — Astronomical techniques (1684) — Convolutional neural networks (1938) — Sky surveys (1464)

1. INTRODUCTION

Weak gravitational lensing infers the matter distribution from coherent $\sim$1% distortions of galaxy shapes. The measurement is limited by knowledge of the PSF: any error in the modeled PSF size or ellipticity propagates directly into the inferred shear. Survey pipelines model the PSF per exposure by fitting that exposure's stars with a flexible model interpolated across the focal plane (PSFEx; E. Bertin 2011) (PIFF; M. Jarvis et al. 2021). This is robust but (i) discards information shared across exposures and nights, (ii) degrades when few clean stars are available, and (iii) is purely empirical at the star positions: it does not exploit auxiliary information such as stellar color, which sets the chromatic PSF.

We take an amortized, learning-based approach. Rather than fit each exposure's PSF from scratch, a single network learns, across many exposures, a *data-adaptive prior* over how the PSF looks and varies, and applies it to a new exposure in one forward pass, mapping that exposure's stars (the *context*) to a continuous PSF field queryable at arbitrary position, color, flux, and resolution. This pools information across the training set that no single-exposure fit can access, and in principle across the focal plane. The inferential target is the PSF at *star-free* positions, where galaxies are measured; predicting held-out stars is the training objective and the validation instrument. We train and evaluate on DES single-epoch data and benchmark against PIFF and PSFEx run by us under matched conditions.

A second difference is the *target* of the model. Traditional PSF models, including PSFEx (super-sampled pixels) and PIFF (a surface-brightness profile on the sky), represent the PSF as a continuous function on $\mathbb{R}^2$ at finer-than-pixel resolution, which must then be convolved with an assumed pixel-response kernel to predict what a detector records. We instead model the *effective* PSF (ePSF; J. Anderson & I. R. King 2000): the instrumental PSF already integrated over the pixel, i.e. the expected fraction of a point source's light falling in each pixel as a function of the source's sub-pixel position. The network maps a query position (and color, flux) directly to these expected pixel values, never representing the continuous PSF explicitly: the sense in which our PSF is *implicit*. Two grounds motivate this target. First, identifiability: from sampled data one cannot uniquely recover the sub-pixel surface brightness without extra assumptions, whereas the ePSF is exactly what the pixels constrain, the reason it underpins state-of-the-art point-source astrometry and photometry (J. Anderson & I. R. King 2000; T. R. Lauer 1999; T. I. Liaudat et al. 2023). Second, sufficiency: the ePSF as a function of source position is a sufficient statistic for forward-modeling *any* source: a galaxy's expected pixel data is its surface brightness convolved with the ePSF, with no separate pixel step (G. M. Bernstein & M. Jarvis 2002), so the ePSF is precisely the kernel a shear pipeline needs. The only residual is discretizing that convolution on a finite grid, which our ability to render at arbitrary oversampling controls directly.

2. METHOD

Architecture—Each detected source is encoded from its pixel cutout (a learned image encoder applied to the scale-normalized stamp), its sky position (sinusoidal encoding), and scalar color and log-flux. A multi-head self-attention block (A. Vaswani et al. 2017) lets every query attend to the context sources; a coordinate decoder then renders the PSF at the query position and any oversampling factor, returning the pixel-convolved effective PSF used to convolve galaxy models. The decoder uses a polar, spin-2-aware angular parametrization, with FiLM modulation by the attended context features.

Inferential target (effective PSF)—The decoder outputs the effective PSF (J. Anderson & I. R. King 2000), the pixel-integrated point-source response, sampled at the query grid, optionally oversampled. Because the loss below compares predicted stamps to *observed* star pixels, the model regresses the observable by construction: it learns $\mathbb{E}[\text{pixel value} \mid \text{point source at } (x, y)]$ directly, rather than fitting a sub-pixel profile and assuming a pixel kernel. Identifiability aligns the three things that a non-identifiable model keeps separate: our training target, the quantity we model, and the quantity we validate all coincide, so reserved-star residuals (§4) directly test what we fit. A surface-brightness model is instead validated only through pixel- or moment-space summaries of its rendered output, the same observable space, so its extra sub-pixel degrees of freedom are never themselves constrained by the data. We score all three methods on moments measured in pixel space (model stamp versus star stamp), so no method is privileged by its internal representation. The same pixel-convolved output is the kernel used to forward-model galaxies in §4.4; oversampling the render reduces the discretization error of that convolution for compact sources.

Loss—Training minimizes an inverse-variance χ^2 between predicted and observed star cutouts, with the per-star amplitude solved in closed form. Star pixels are noisy realizations whose conditional mean is the effective PSF; a squared-error objective is minimized by the conditional expectation $\mathbb{E}[\,\text{pixel} \mid \text{source, context}\,]$, so with enough capacity and data the objective is minimized by that quantity, the identifiable effective PSF. The network thus targets only what the pixels constrain, rather than a finer-than-pixel profile the sampled data do not determine. We introduce a *blend forward-model* loss: rather than treat each clean star as isolated, we model its cutout as a generalized-least-squares sum of the target plus its detected neighbors, all rendered by the same decoder (§4.7). Detected galaxies near a target are handled by exclusion (default), masking, or a crude extended-component model; the three are within noise on real reserved stars.

Point-source context—The PSF should be informed only by point sources. We restrict the attention context to stellar detections; §4.7 shows this is necessary to avoid a spin-2 corruption from extended sources in the context.

Color conditioning—The decoder conditions on $g{-}i$ color, enabling a per-object chromatic PSF (differential chromatic refraction and wavelength-dependent seeing) that per-exposure empirical models cannot represent (§4.5).

2.1. *Contamination correction by Monte Carlo EM*

Reserved "clean" stars are never truly clean: sub-threshold blends add faint light to their cutouts and bias the learned PSF toward an under-concentrated profile. We remove this bias with a latent-variable model fit by expectation maximization (A. P. Dempster et al. 1977).

Model—For clean star i, the cutout $\mathbf{y}_i$ is a bright central source — the star, whose surface brightness is the

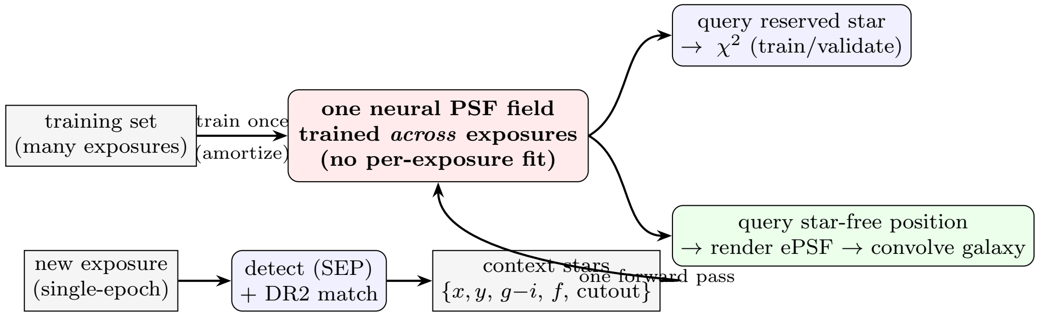

Figure 1. Overall workflow and amortization. A single network is trained once across many exposures (top) and then applied to a new exposure in one forward pass, with no per-exposure fitting: its detected stars (context) are mapped to a continuous PSF field (bottom). The *same* field is queried two ways: at held-out reserved stars, giving the training objective and validation, and at star-free positions, where the rendered (oversampled) effective PSF convolves galaxy models—the inferential target.

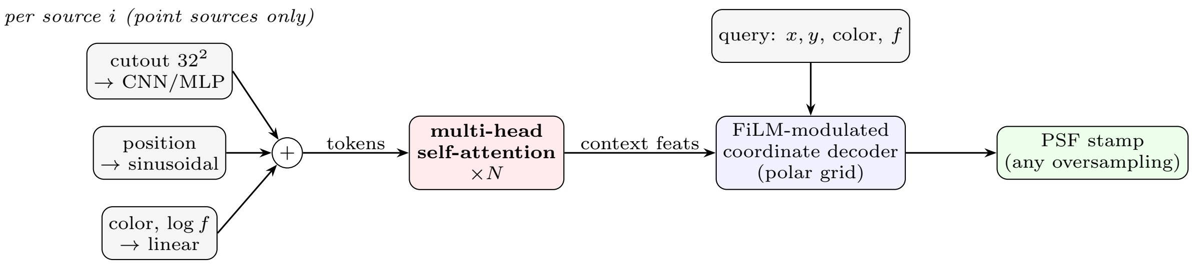

Figure 2. Network architecture. Each context source (restricted to point sources; §2) is encoded from its pixel cutout, sky position, and scalar color/log-flux into a token. Multi-head self-attention mixes the context; a FiLM-modulated coordinate decoder, conditioned on the attended features and the query (position, color, flux), renders the pixel-convolved PSF at any position and oversampling factor.

effective PSF $\boldsymbol{\pi}_\theta(\mathbf{x}_i)$ scaled by an unknown amplitude a_i — plus faint nuisance sources and noise with the DES inverse-variance weights $\mathbf{w}_i$,

$$\mathbf{y}_i = a_i\,\boldsymbol{\pi}_\theta(\mathbf{x}_i) + \textstyle\sum_j f_{ij}\,\mathbf{g}(\boldsymbol{\mu}_{ij}, \Sigma_{ij}) + \boldsymbol{\varepsilon}_i. \qquad (1)$$

The nuisance sources are too faint to classify, so we model each as a single bivariate Gaussian $\mathbf{g}(\boldsymbol{\mu}, \Sigma)$ of flux f, sub-pixel centroid $\boldsymbol{\mu}$, and covariance $\Sigma = \Sigma_\theta(\mathbf{x}_i) + LL^\top$, where Σ_θ is the current PSF second-moment matrix and $LL^\top \succeq 0$ (Cholesky factor L) is a non-negative intrinsic broadening. A contaminant is thus at least as broad as a point source and may be extended and elliptical, with no star/galaxy dichotomy.

Latents, prior, and parameters—The nuisance configuration $\mathbf{z}_i = \{(b_{ic}, f_{ic}, \boldsymbol{\mu}_{ic}, L_{ic})\}$ lives on a per-pixel detection grid, at most one source per cell ($b_{ic} \in \{0, 1\}$): a fixed-dimensional mean-field representation that avoids transdimensional moves. It carries an informative, physically motivated prior $p(\mathbf{z})$ — per-cell Bernoulli rate, truncated power-law flux (slope and bright cutoff set by DES source counts and the detection limit that defines "sub-threshold"), and a weak prior on L. This prior alone resolves the degeneracy between a broad PSF with no contamination and a sharp PSF with faint contamination, and so sets the magnitude of the correction. Crucially, its hyperparameters — the per-cell detection rate λ and the flux power-law slope α — are unknown on real data, so we do not fix them: they are *global* random variables shared across stars, with weakly-informative hyperpriors (a conjugate Gamma rate; a Gamma slope), and are inferred jointly. The contamination level is thus learned from the many-star aggregate rather than assumed; §4.7 tests on simulations with a known PSF field whether the inferred (λ, α) recover the generator's together with the PSF concentration and galaxy sizes. We place *no* prior on the network weights θ — we seek the maximum marginal-likelihood PSF, not a Bayesian network — and none on the central amplitudes a_i. These are parameters: a_i is profiled in closed form and θ is fit in the M-step. Only the counts-endowed nuisances are marginalized,

$$\hat{\theta} = \arg\max_\theta \sum_i \log \int p(\mathbf{y}_i \mid \theta, \mathbf{z}_i)\, p(\mathbf{z}_i)\, d\mathbf{z}_i, \qquad (2)$$

with a_i profiled inside the integrand.

Monte Carlo EM—The integral is intractable, so we use EM with the E-step expectation replaced by a Monte Carlo average over contaminant draws (G. C. G. Wei & M. A. Tanner 1990). At iteration t with current PSF $\theta^{(t)}$: (i) *E-step* — for each star draw K samples

$\mathbf{z}_i^{(t,k)} \sim p(\mathbf{z}_i \mid \mathbf{y}_i, \theta^{(t)})$ by collapsed Gibbs, profiling a_i each sweep (it has no prior), marginalizing each flux in closed form, resampling detection, centroid, and covariance per cell, and — globally across all stars — resampling (λ, α) from the pooled contaminant sufficient statistics, a hierarchical Gibbs step; the Monte Carlo objective is $\hat{Q}(\theta \mid \theta^{(t)}) = K^{-1} \sum_{i,k} \log p(\mathbf{y}_i \mid \theta, \mathbf{z}_i^{(t,k)})$. (ii) *M-step* — increase $\hat{Q}$ by stochastic gradient descent warm-started from $\theta^{(t)}$, a generalized EM step (A. P. Dempster et al. 1977) since the network is not maximized exactly; operationally each minibatch draws one of the K imputations per star and subtracts it, training against the multiply imputed cleaned stars so the contaminant posterior *uncertainty* propagates into the fit rather than a point estimate. (iii) Iterate until the PSF and the recovered galaxy sizes (§4.7) stop changing; later iterations' sharper central template removes residual contamination that the contaminated initial template leaves. We increase K across iterations and monitor Monte Carlo error (J. G. Booth & J. P. Hobert 1999; R. A. Levine & G. Casella 2001); a stochastic-approximation variant (B. Delyon et al. 1999) amortizes the per-iteration resampling cost.

Sampler validation—Two checks gate the E-step. *Calibration*: simulation-based calibration — draw $\mathbf{z}$ from the prior, synthesize a stamp, infer, and require nominal credible-interval coverage of the total contaminant flux and count (this caught three sampler bugs in development). *Mixing*: per star we run dispersed-initialization chains and report the Gelman–Rubin $\hat{R}$ and effective sample size of the flux and count summaries; the E-step proceeds only where $\hat{R} \approx 1$ with adequate effective sample size, so a poorly mixing chain cannot silently corrupt the fit.

3. DATA

We use DES single-epoch FITS (Y5A1, CCD 31; amplifier B is dead, masked via MSK bit 8). Sources are detected with SEP and cross-matched to DES DR2 for colors, `SPREAD_MODEL` star/galaxy labels, and coadd object IDs. Splits are at the observing-night level by deterministic hash to prevent leakage (consecutive exposures share atmosphere), frozen in a committed manifest with ∼20% of clean stars per test/validation exposure reserved for scoring only; the test split is 3,301 exposures across $grizY$. Reserved stars are excluded from every method's fit/context and scored identically across methods on the same pixel grid.

Simulations—A matched pipeline renders Moffat PSF fields into the same data format, with FWHM and shear varying as second-order polynomials across the field. This variation lies within the polynomial spatial model that PIFF and PSFEx fit, so it does not handicap the baselines. The renderer optionally adds a chromatic dependence of FWHM on color, blended stars, and injected galaxy detections. Star density, galaxy density (3:1), and per-source S/N are tuned to match the real extraction. The simulation is the controlled instrument for the systematics studies in §4.7.

4. RESULTS

4.1. *Reserved-star accuracy on real data*

On 3,301 test exposures and 73,609 reserved stars, our model (no per-exposure fitting) matches or exceeds both baselines on every metric (Table 1). Every method is scored on the identical set of reserved stars.

4.2. *Pixel-level and astrometric validation*

Because the model regresses the effective PSF (the expected pixel values), the reserved-star quality of fit can be assessed directly in the modeled (pixel) space, the validation native to effective-PSF work (J. Anderson & I. R. King 2000). Two checks follow. First, the inverse-variance χ^2/dof is a noise-calibrated goodness of fit: at 1.04 (Table 1) the per-pixel residuals are consistent with photon and read noise, i.e. no unmodeled pixel-level structure, a statement available to a sub-pixel model only through its rendered output. The variance is the survey's own inverse-weight (photon plus read noise) map, not a quantity tuned to the residuals, so $\chi^2 \approx 1$ is an independent calibration check rather than a fitted outcome. Second, the predicted ePSF centroid matches the star centroid to 0.010 px (2.6 mas) in median offset, versus 0.017 px (4.5 mas) for PIFF and 0.020 px (5.1 mas) for PSFEx (Table 2); all three centroids are measured with the same adaptive-moment routine on the same pixel grid, so the difference reflects the models, not a centroiding convention. This is a near 2× improvement on a quantity the ePSF formalism is built to optimize.

Third, stacking the unit-normalized residual (observed star minus model, as a fraction of total light) over 306 reserved stars isolates any coherent shape error that second moments can miss (Figure 6). Our model leaves the flattest residual (RMS 2.6×10^{-4}); PIFF and PSFEx leave a visible negative core (3.5 and 4.4×10^{-4}), a higher-order mismatch in the central pixels that is consistent with their larger size residuals and is invisible to the size and ellipticity summaries alone.

4.3. *Residual two-point statistics*

Per-star residuals understate the impact on weak lensing, where the *spatially coherent* part of the PSF-

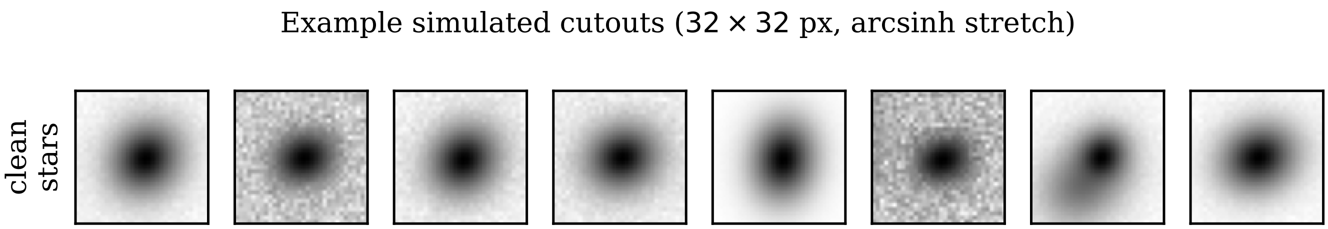

Figure 3. Example simulated clean-star cutouts (32×32 px, arcsinh stretch): the compact point sources that inform the PSF. Galaxy detections also populate the simulated fields but, under the point-source-context policy (§2, §4.7), are excluded from the attention context.

Table 1. Reserved-star residuals on the DES test split (3,301 exposures, 73,609 stars). T scatter is the robust standard deviation of $\delta T/T$; $|\delta e|$ are median absolute ellipticity residuals; corr is the model–star shape correlation. Best in bold.

Method	T scat	$\lvert\delta e_1\rvert$	$\lvert\delta e_2\rvert$	corr(e_1)	corr(e_2)	χ^2/dof
Neural PSF (ours)	**0.0301**	**0.0151**	**0.0150**	**0.650**	**0.607**	1.037
PIFF	0.0393	0.0171	0.0158	0.611	0.577	1.073
PSFEx	0.0371	0.0160	0.0154	0.645	0.604	1.047

Table 2. Astrometric residual of the model PSF centroid relative to the star, on the reserved test stars. Per-axis scatter is the robust standard deviation; the offset is the median radius ($0.263''$/px). Best in bold.

Method	$\sigma(\delta x)$ [px]	$\sigma(\delta y)$ [px]	median $\lvert\delta\rvert$ [mas]
Neural PSF (ours)	**0.008**	**0.009**	**2.6**
PIFF	0.015	0.013	4.5
PSFEx	0.015	0.017	5.1

modeling error sources additive shear bias. We therefore compute the standard ρ-statistics (B. Rowe 2010; M. Jarvis et al. 2016) of the reserved-star residuals (the auto- and cross-correlations of the residual ellipticity δe and the size-leakage term $e\,\delta_T$), accumulated within exposures with TREECORR (Figure 7). On the same reserved stars used for Table 1, our model is at or below both per-exposure baselines on four of the five statistics: the median $|\xi_+|$ across angular scales is 3.9×10^{-5} (ρ_1), 5.2×10^{-5} (ρ_2), 2.5×10^{-6} (ρ_3), and 4.8×10^{-6} (ρ_4), against $\rho_1, \rho_2 \gtrsim 7 \times 10^{-5}$ and $\rho_3 \simeq 6 \times 10^{-5}$ for PIFF. The ρ_3 result, more than an order of magnitude below PIFF, reflects the model's smaller and less spatially structured size residuals. The one exception is $\rho_5 = \langle e\,(e\,\delta_T)\rangle$, where our model ($5.4 \times 10^{-5}$) sits above the baselines ($\sim 10^{-5}$): a residual coupling between PSF shape and the size error that a weak-lensing analysis would carry in its additive-bias budget. Overall the coherent, lensing-relevant part of the residual is no larger than that of the per-exposure baselines, despite our model fitting no per-exposure free parameters.

4.4. *Galaxy recovery on real data*

We test the inferential goal directly. We inject GALSIM galaxies, convolved with a strictly isolated reserved star's empirical PSF, at that star's position; each method predicts the PSF there from the *other* stars and recovers the galaxy. Recovering with the true PSF (the anchor's own image) sets the protocol floor: near-zero ellipticity bias and a residual -1.2% size bias—not a true zero, but the discretization floor of the recovery forward model—against which the predicted PSFs are scored. On 32 r-band test exposures (170 anchors; Table 3), all three methods recover galaxy *ellipticity* with low bias

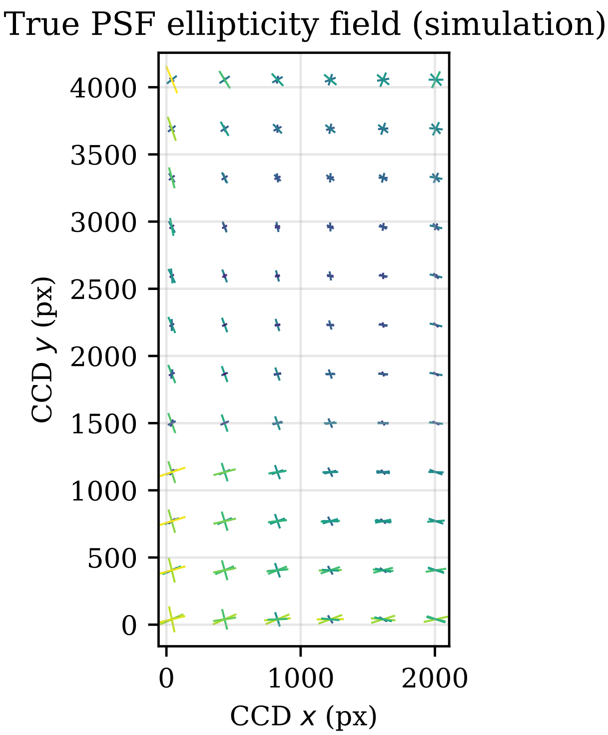

Figure 4. The true, spatially varying PSF ellipticity field in a simulated exposure (whisker length $\propto |e|$); the network must recover this at star-free positions.

($|\Delta e_1| \lesssim 0.013$): the pixel-convolved effective PSF carries the PSF ellipticity faithfully into the forward model, so no method imprints a coherent shape error on the recovered galaxies, the weak-lensing-critical property, which our model satisfies on par with the per-exposure baselines. The methods separate instead on recovered *size*: PIFF is slightly large (+1.1%), while our recovered sizes are ∼8% small. This is *not* a second-moment size error (rendered at a fiducial flux, the model's PSF size is accurate at both star and star-free positions, within 1%), but a profile-shape effect: the model's PSF is slightly *under-concentrated*, carrying ∼2% less encircled energy in the central few pixels than the empirical PSF, which biases recovered galaxy sizes low. The magnitude is architecture-dependent (Table 3 reports the baseline; higher-capacity coordinate encodings—random Fourier features, an explicit analytic core—roughly halve it, §5), pointing to an under-resolution component. It is, however, specific to real data: controlled simulations with known PSFs do not reproduce it at any decoder capacity, so it is not a generic spectral-bias limitation of the decoder. Its full origin is not yet isolated: contamination of nominally-clean training stars by faint unresolved neighbours reproduces the sign and PIFF's relative robustness but only a fraction of the magnitude, and unmodeled real PSF morphology is the leading remaining candidate. The size deficit is the main residual systematic of our model; recovered ellipticity, the consequential quantity for shear, is unbiased.

Table 3. Star-anchored galaxy injection–recovery (32 r-band test exposures, 170 anchors). The true-PSF recovery sets the protocol floor. All methods recover ellipticity with low bias; they separate on recovered size.

PSF	Size bias	Δe_1	Δe_2
True PSF	−1.2%	−0.009	−0.000
Neural PSF (ours)	−7.6%	−0.005	−0.000
PIFF	+1.1%	−0.013	−0.003

4.5. *Chromatic PSF correction*

In a chromatic simulation (FWHM varies −3%/mag of $g-i$), color conditioning reduces the residual chromatic size slope from −0.06/mag (PIFF, PSFEx, and our own zero-color ablation) to +0.004/mag, a ∼15× reduction. On real r-band data the paired color-versus-zero-color slope difference is +0.0024/mag (CI $[+0.0016, +0.0030]$); the baselines retain the chromatic slope because a per-exposure empirical PSF has no access to per-star color.

4.6. *Sample efficiency*

Restricting every method to k randomly chosen fit stars, our model is approximately flat in k (size scatter $0.078 \to 0.056$, $\chi^2 \sim 1.1$ from $k = 5$ to $k = 100$), while PIFF collapses below $k \approx 25$ ($\chi^2 = 5.1$, scatter 0.18 at $k = 5$). Amortization transfers information from the training set to sparsely-populated exposures.

4.7. *Simulation studies*

Architecture ladder—On the star-free truth grid, χ^2/dof improves along additive → FiLM → diagonal → polar ($8.9 \to 4.96 \to 4.41$; PIFF 2.67; verified floor 1.0), recovering both ellipticity components.

Blend forward-model—On a blended simulation, single-star training leaves a +4.8% size bias on the star-free truth grid; the blend forward-model loss reduces it to +0.7% (≈7×).

Galaxy-context corrupts PSF ellipticity—and the fix—Injecting galaxy detections into the simulation, a model that attends to *all* detections (including galaxies) suffers a clean, reproducible failure: it cannot represent one spin-2 ellipticity component [corr(e_1) = 0.05 under

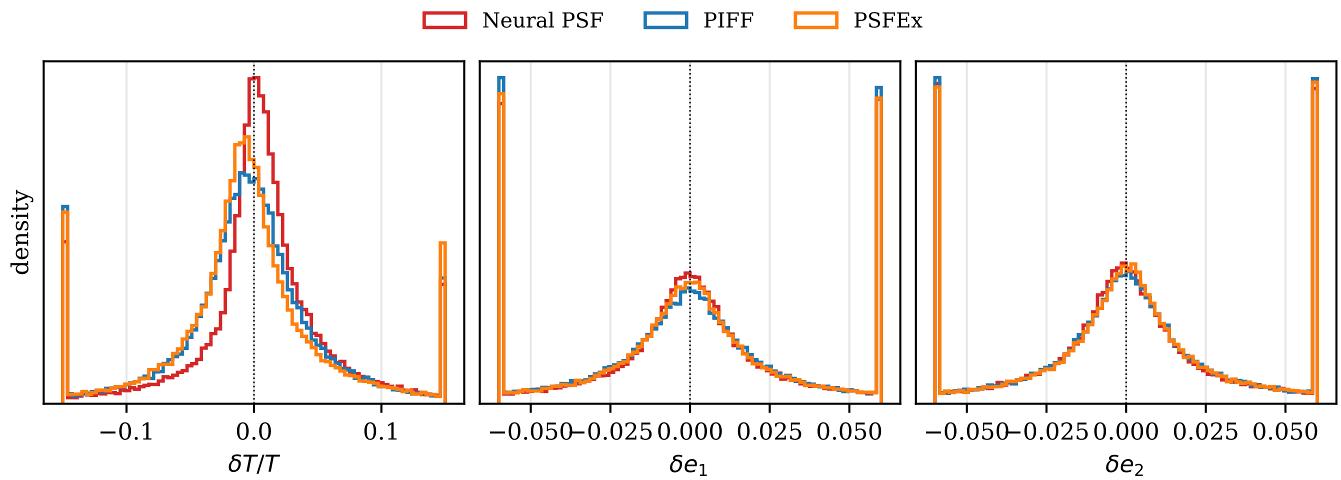

Figure 5. Reserved-star residual distributions on the DES test split. Our model (red) is the narrowest in size and ellipticity.

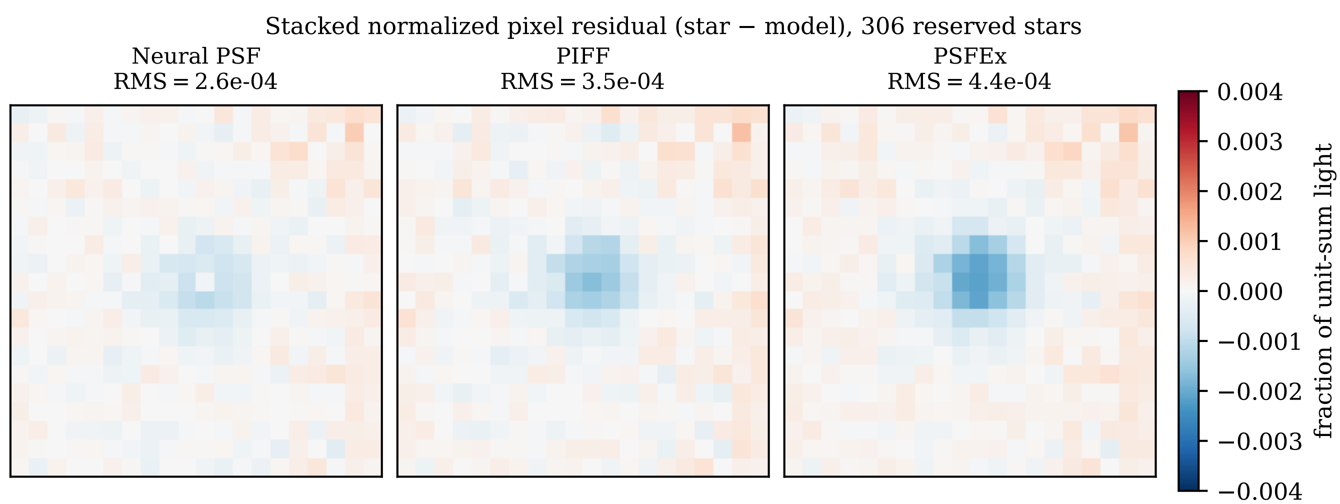

Figure 6. Stacked normalized pixel residual (observed star − model, unit-sum) over 306 reserved test stars, on a common scale. Our model leaves the flattest residual; the per-exposure baselines retain a coherent central core—a higher-order shape error that second-moment metrics do not capture (§4.2).

the polar parametrization; the failure flips to e_2 under a Cartesian parametrization, identifying a representation degeneracy rather than noise], while recovering the other perfectly. We localize the cause by ablation: it is *not* data volume, per-source S/N, galaxy size or color, or the χ^2 cap—the model is treating extended galaxy cutouts as PSF evidence. Restricting the attention context to point sources recovers *both* components completely [$\mathrm{corr}(e_1, e_2) = 0.98$, matching the galaxy-free ideal]; a coordinate-system change does not. Galaxies are clearly distinguishable (cutout second moment $T \approx 33$ versus $15\,\mathrm{px}^2$ for stars), so this is the model failing to use available information, fixed by the principled restriction—which we adopt as the production policy—that only point sources inform a PSF. Excluding galaxies from the context costs nothing measurable on the real test set: reserved-star residuals are unchanged from the all-detections ablation (size scatter 0.031 versus 0.030; identical median ellipticity error and shape correlation), so the policy removes the simulation failure mode without trading away real-world performance.

Seed regularization: EMA versus ensembling—Because the field is learned once and applied without per-exposure refitting, it carries seed-to-seed prediction noise. On the star-free truth grid we compare a single model, an exponential moving average (EMA) of one model's weights, and a 4-seed deep ensemble (averaging predicted stamps), each over matched 120-epoch runs (Table 4). EMA gives a small, consistent gain (T-scatter

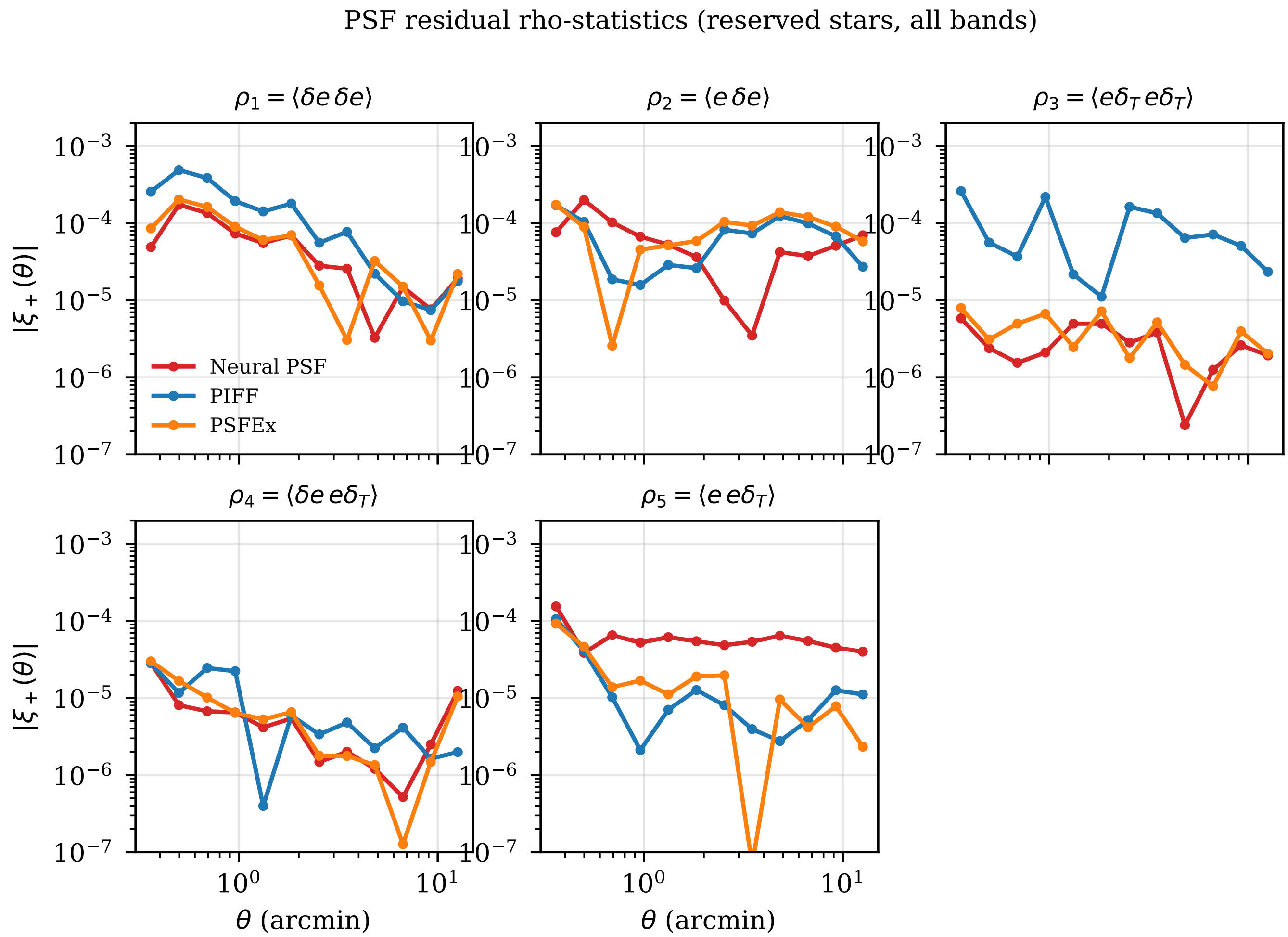

Figure 7. PSF residual ρ-statistics on reserved stars, per method. Our model (red) is at or below both baselines on ρ_1–ρ_4; only ρ_5 (the shape–size coupling) is modestly larger (§4.3).

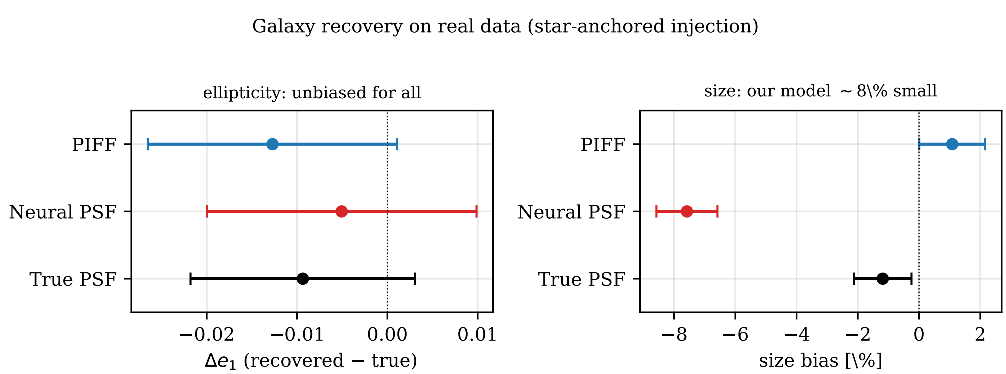

Figure 8. Galaxy recovery on real data (star-anchored injection), median ± bootstrap 95% CI per PSF. Left: recovered ellipticity is low-bias ($|\Delta e_1| \lesssim 0.013$) for all three methods. Right: the methods separate on recovered *size*—our model runs ~8% small (the PSF-core under-concentration), while PIFF and the true PSF are near zero (§4.4).

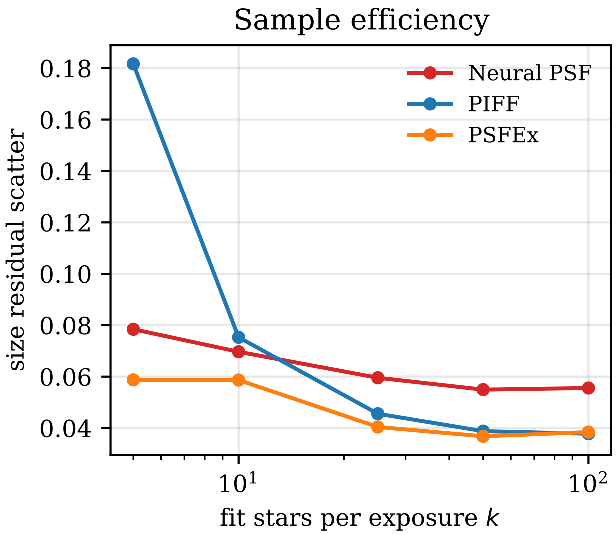

Figure 9. Size-residual scatter versus the number of fit stars k. Our model is nearly flat in k; PIFF degrades sharply below $k \approx 25$.

Table 4. Truth-grid PSF recovery (4 matched 120-epoch seeds): seed-noise scatter for a single model, EMA, and a 4-seed ensemble.

	$\sigma(\delta e_1)$	$\sigma(\delta e_2)$	$\sigma(\delta T/T)$
single model	0.0094	0.0099	0.0175
EMA (1×)	0.0092	0.0096	0.0167
ensemble (4×)	**0.0073**	**0.0075**	**0.0154**

−5%, ellipticity $\lesssim$ 3%); the ensemble is much larger (−12% size, −24% ellipticity). EMA is therefore *not* a substitute for ensembling: it captures only a fraction of the benefit. The per-seed aggregate scatter is itself small (the model is seed-stable), but the per-position predictions are diverse, which only ensembling exploits. On real reserved stars the ensemble improves the metrics by only ∼1%: that residual is measurement-noise- and blend-limited (corr∼0.66, not the truth-grid 0.99), so it masks the model-error reduction that ensembling delivers at the star-free positions where galaxies are measured.

5. DISCUSSION

Our amortized approach trades a per-exposure fit for a single trained model. Benefits: no test-time fitting, robustness in star-sparse exposures, conditioning on color (chromatic PSF) and flux (brighter–fatter), and a directly forward-modelable output: we predict the pixel-convolved effective PSF, the identifiable quantity the data constrain and the exact kernel a galaxy model is convolved with (§1), so the PSF feeds a shear pipeline with no intervening pixel-deconvolution assumption. Caveats: (i) unlike per-exposure fitters, our approach requires many exposures to learn the field (in simulation, ellipticity recovery is unstable below a few thousand exposures; real DES is well above this); (ii) the simulation's idealized PSF (Moffat, low-order polynomial field) favors the per-exposure baselines, so absolute-accuracy comparisons rest on real data; and (iii) the galaxy-context finding argues that a PSF estimator's context should be gated to point sources, which we adopt. (iv) The model slightly *under-concentrates* the PSF core: rendered at a fiducial flux its second-moment size is accurate, but it carries ∼2% less encircled energy in the central few pixels than the empirical PSF, which biases recovered galaxy sizes a few percent low (§4.4). Higher-capacity coordinate encodings reduce it: widening the decoder's Fourier frequency basis cuts the encircled-energy residual at r=2 px (∼2% → 0.1%) and the galaxy-size bias by ∼30–50%, and an explicit analytic core does likewise, indicating an under-resolution component. It is, however, *specific to real data*: controlled simulations with known PSFs do not reproduce the deficit at any decoder capacity, ruling out a generic spectral-bias limitation of the multilayer-perceptron decoder. Its full origin is under investigation—faint-neighbour contamination of the training stars reproduces the sign and PIFF's robustness but, in simulation, only a fraction of the magnitude and without a clear dose-response, so unmodeled real PSF morphology is the leading remaining candidate. It is the main residual systematic to address before a shear analysis. A separate, milder effect appears only in the reserved-star metric (Fig. 14): evaluated at each star's *own* flux, $\delta T/T$ rises to ∼+1.2% at faint magnitudes, a spurious flux–size trend in the conditioning; because the inferential PSF is rendered at a fiducial flux, this does not propagate to galaxy measurement.

6. CONCLUSION

A single attention-based continuous PSF field, trained across DES single-epoch exposures and applied with no per-exposure fitting, matches or exceeds PIFF and PSFEx on real reserved stars and recovers galaxy ellipticity without coherent bias (on par with PIFF) on a star-anchored injection test, while additionally correcting a chromatic systematic and remaining robust in star-sparse exposures. Controlled simulations motivate two training choices: a blend forward-model loss and point-source-only attention context. The latter follows from

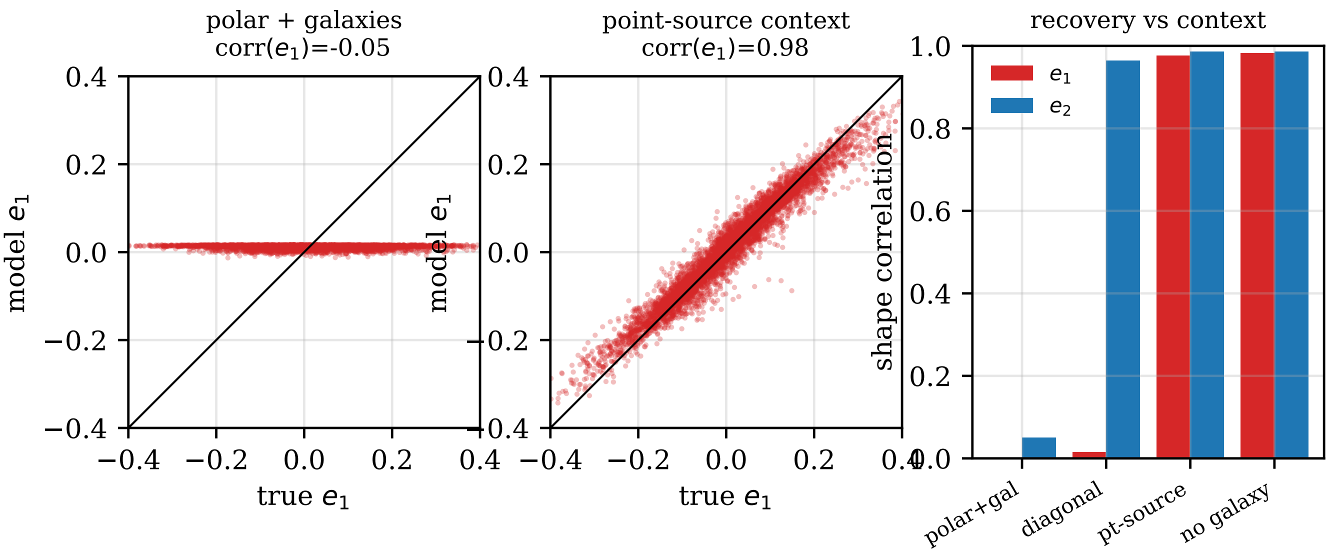

Figure 10. Galaxy detections in the attention context corrupt the recovered PSF ellipticity. Left: with galaxies in context, model e_1 collapses to a constant (corr$= -0.04$). Middle: restricting the context to point sources recovers it (corr$= 0.98$). Right: shape correlations across configurations—point-source context and galaxy-free both recover *both* components; a coordinate change (diagonal) does not.

diagnosing a spin-2 representation failure triggered by galaxy detections in context.

ACKNOWLEDGMENTS

This material is based on work supported by the National Science Foundation under Grant No. 2209720 and the U.S. Department of Energy, Office of Science, Office of High Energy Physics under Award Number DE-SC0023714.

This paper has undergone internal review in the LSST Dark Energy Science Collaboration.

The DESC acknowledges ongoing support from the Institut National de Physique Nucléaire et de Physique des Particules in France; the Science & Technology Facilities Council in the United Kingdom; and the Department of Energy and the LSST Discovery Alliance in the United States. DESC uses resources of the IN2P3 Computing Center (CC-IN2P3–Lyon/Villeurbanne - France) funded by the Centre National de la Recherche Scientifique; the National Energy Research Scientific Computing Center, a DOE Office of Science User Facility supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231; STFC DiRAC HPC Facilities, funded by UK BEIS National E-infrastructure capital grants; and the UK particle physics grid, supported by the GridPP Collaboration. This work was performed in part under DOE Contract DE-AC02-76SF00515.

This project used public archival data from the Dark Energy Survey (DES). Funding for the DES Projects has been provided by the U.S. Department of Energy, the U.S. National Science Foundation, the Ministry of Science and Education of Spain, the Science and Technology FacilitiesCouncil of the United Kingdom, the Higher Education Funding Council for England, the National Center for Supercomputing Applications at the University of Illinois at Urbana-Champaign, the Kavli Institute of Cosmological Physics at the University of Chicago, the Center for Cosmology and Astro-Particle Physics at the Ohio State University, the Mitchell Institute for Fundamental Physics and Astronomy at Texas A&M University, Financiadora de Estudos e Projetos, Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro, Conselho Nacional de Desenvolvimento Científico e Tecnológico and the Ministério da Ciência, Tecnologia e Inovação, the Deutsche Forschungsgemeinschaft, and the Collaborating Institutions in the Dark Energy Survey.

The Collaborating Institutions are Argonne National Laboratory, the University of California at Santa Cruz, the University of Cambridge, Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas-Madrid, the University of Chicago, University College London, the DES-Brazil Consortium, the University of Edinburgh, the Eidgenössische Technische Hochschule (ETH) Zürich, Fermi National Accelerator Laboratory, the University of Illinois at Urbana-Champaign, the Institut de Ciències de l'Espai (IEEC/CSIC), the Institut de Física d'Altes Energies, Lawrence Berkeley National Laboratory, the Ludwig-Maximilians Universität München and the associated Excellence Cluster Universe, the University of Michigan, the National Optical Astronomy Observatory, the University of Nottingham, The Ohio State University, the OzDES Membership Consortium, the University of Pennsylvania, the University of Portsmouth, SLAC National Accelerator Laboratory, Stanford University, the University of Sussex, and Texas A&M University.

Based in part on observations at Cerro Tololo Inter-American Observatory, National Optical Astronomy Observatory, which is operated by the Association of Universities for Research in Astronomy (AURA) under a cooperative agreement with the National Science Foundation.

Baselines: PIFF (PixelGrid, second-order `BasisPolynomial`, 4σ rejection) and PSFEx (`PSF_SAMPLING` 0.5, 45×45, `PSFVAR_DEGREES` 2; rendered via GALSIM `DES_PSFEx`, `method='no_pixel'`).

Software: Astropy (Astropy Collaboration 2022), GalSim (B. T. P. Rowe et al. 2015), SEP (K. Barbary 2016), PyTorch (A. Paszke et al. 2019), PIFF (M. Jarvis et al. 2021), PSFEx (E. Bertin 2011)

APPENDIX

A. REAL-DATA STAR SELECTION AND PSF EXAMPLES

Figure 11 shows a region of a real DES single-epoch r-band exposure with the source selection overlaid: clean stars used for fitting and scoring (green), held-out reserved stars (blue), and additional context-only stellar detections (orange). Galaxies (the extended sources, e.g. the bright object near the center) are detected but excluded from the attention context (§2). Figure 12 compares, for a single high-S/N reserved test star, the observed stamp against the PSFEx, PIFF, and Neural PSF models evaluated at that position (each method having seen only the non-reserved stars). All three reproduce the stellar profile; PIFF's per-exposure interpolation is visibly noisier in the wings.

B. BENCHMARK VALIDATION DIAGNOSTICS

For comparison with standard per-exposure PSF validation (cf. M. Jarvis et al. 2021), we report the same diagnostics. Table 5 breaks the metrics down by band. Our model has the smallest size scatter and median ellipticity error in every band.

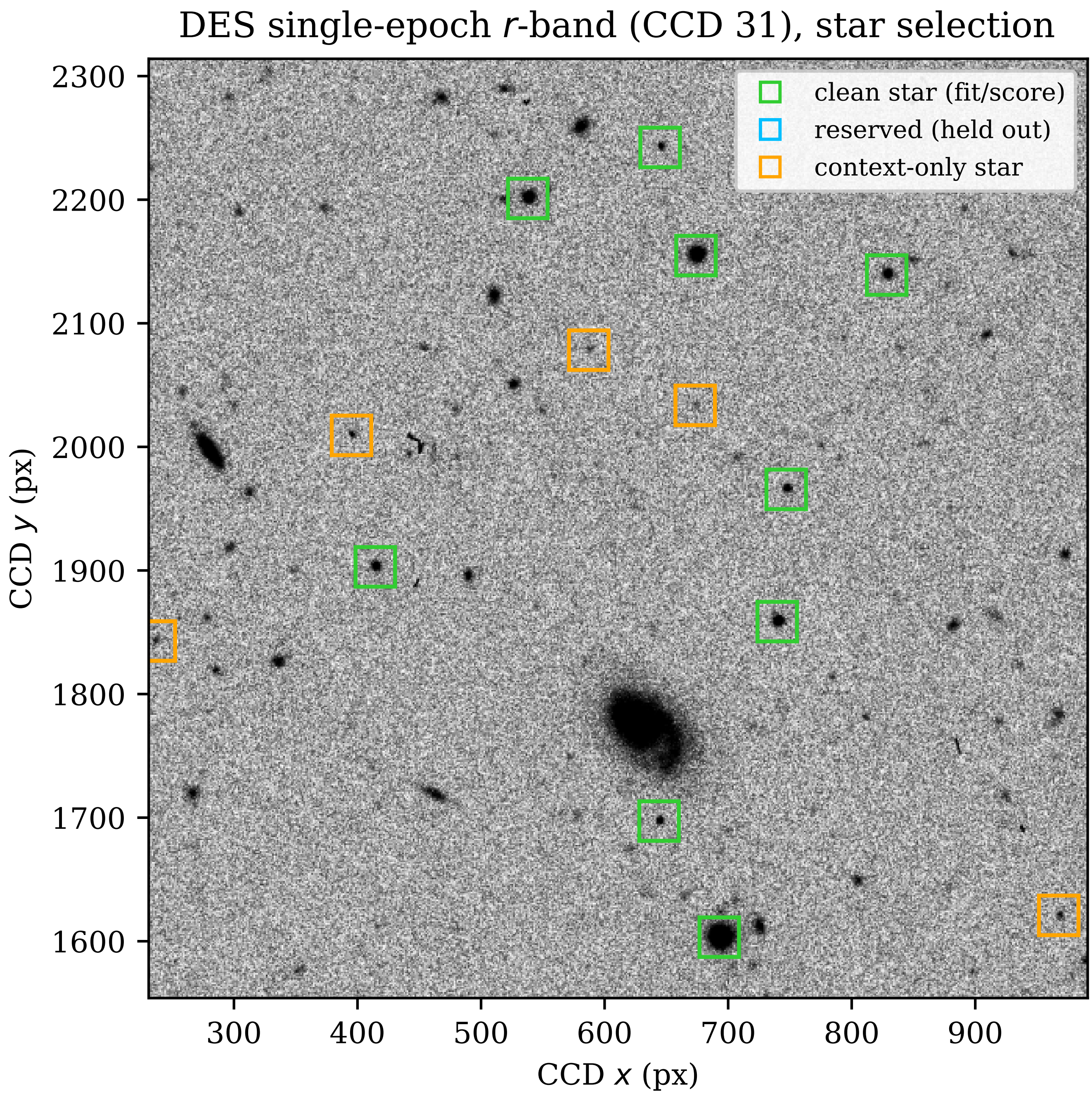

Figure 11. Star selection on a real DES exposure. Green: clean stars (fit and score). Blue: reserved (held out for evaluation). Orange: context-only stellar detections. Extended sources (galaxies) are excluded from the attention context.

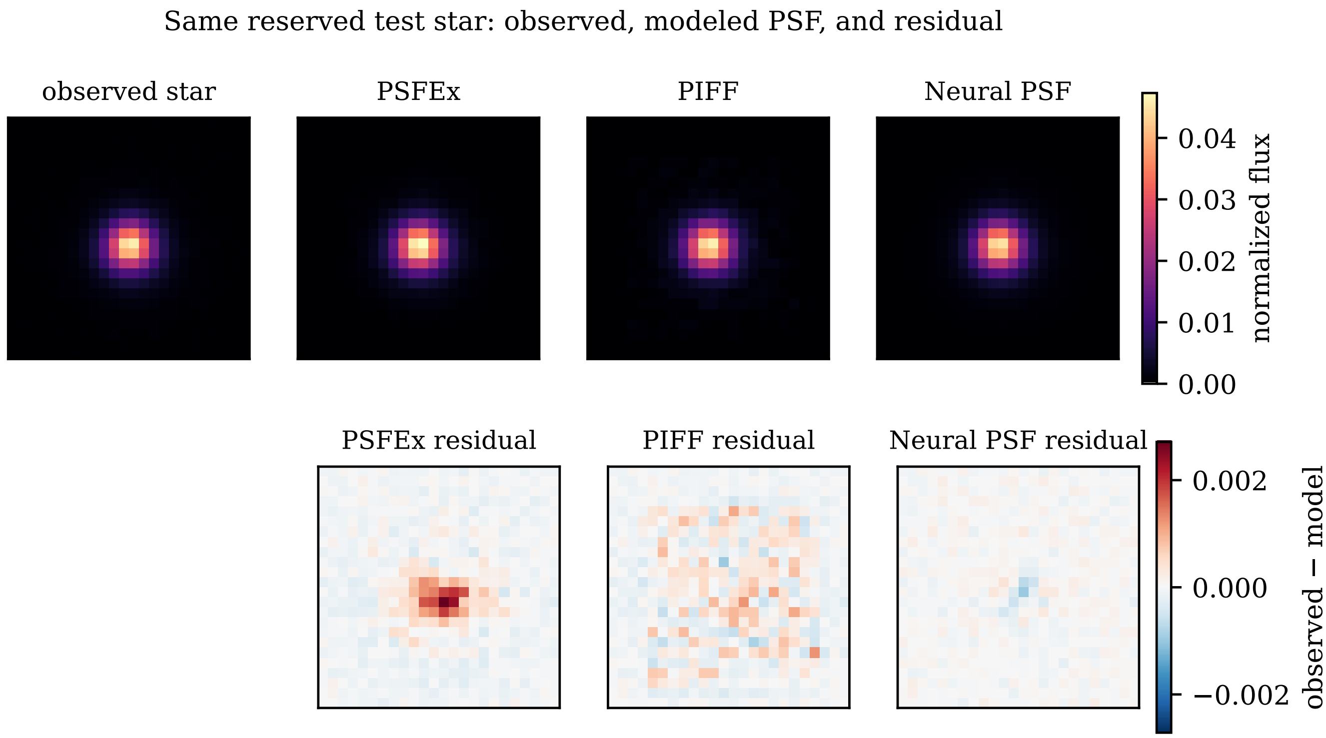

Figure 12. Top: the same reserved test star (left, observed) and the PSF predicted at its position by PSFEx, PIFF, and Neural PSF, on a common linear stretch. Bottom: the residual (observed − model) for each method, on a common diverging scale. Each model was fit/conditioned on the non-reserved stars only.

Table 5. Reserved-star residuals by band (test split). T scatter is the robust standard deviation of $\delta T/T$; $|\delta e_1|$ is the median absolute e_1 residual; corr is the model–star e_1 correlation. Best per band in bold.

Band	Method	T scat	$\lvert\delta e_1\rvert$	corr(e_1)
g	**Neural PSF (ours)**	**0.0331**	**0.0164**	**0.475**
	PIFF	0.0439	0.0188	0.448
	PSFEx	0.0417	0.0173	0.473
r	**Neural PSF (ours)**	**0.0291**	**0.0144**	**0.576**
	PIFF	0.0366	0.0164	0.535
	PSFEx	0.0340	0.0151	0.572
i	**Neural PSF (ours)**	**0.0274**	**0.0136**	**0.643**
	PIFF	0.0342	0.0155	0.605
	PSFEx	0.0339	0.0144	0.637
z	**Neural PSF (ours)**	**0.0272**	**0.0140**	**0.601**
	PIFF	0.0387	0.0160	0.503
	PSFEx	0.0359	0.0152	0.589
Y	**Neural PSF (ours)**	**0.0391**	**0.0210**	**0.568**
	PIFF	0.0513	0.0226	0.523
	PSFEx	0.0449	0.0221	0.550

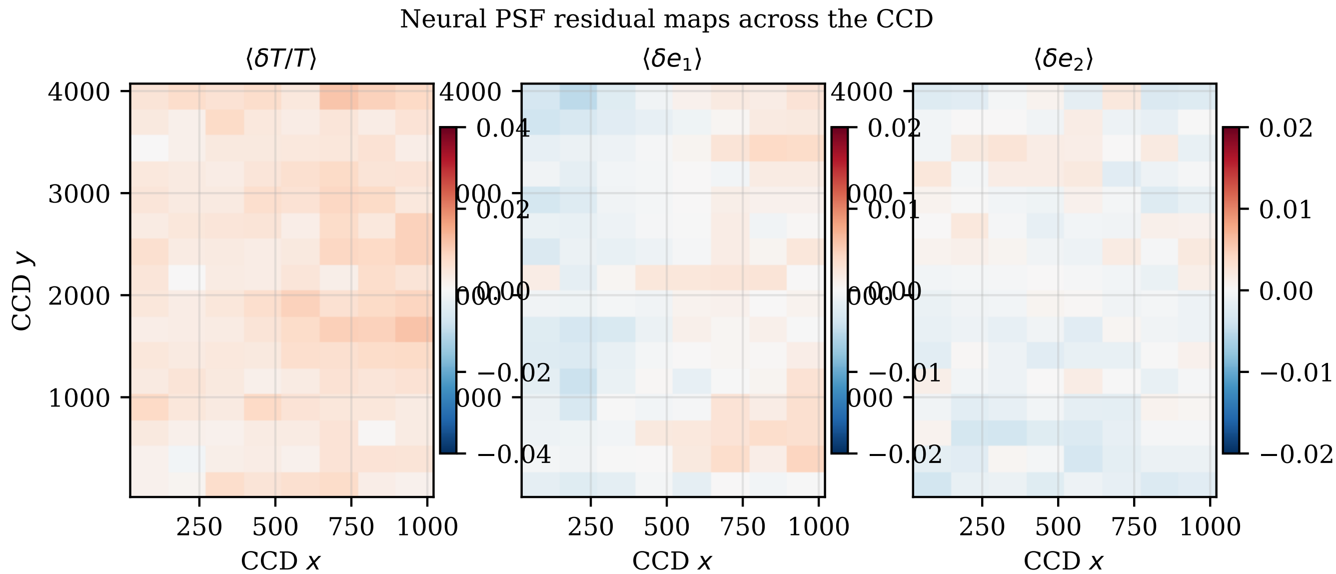

Figure 13. Median Neural PSF residuals binned across the CCD (cf. PIFF Fig. 9).

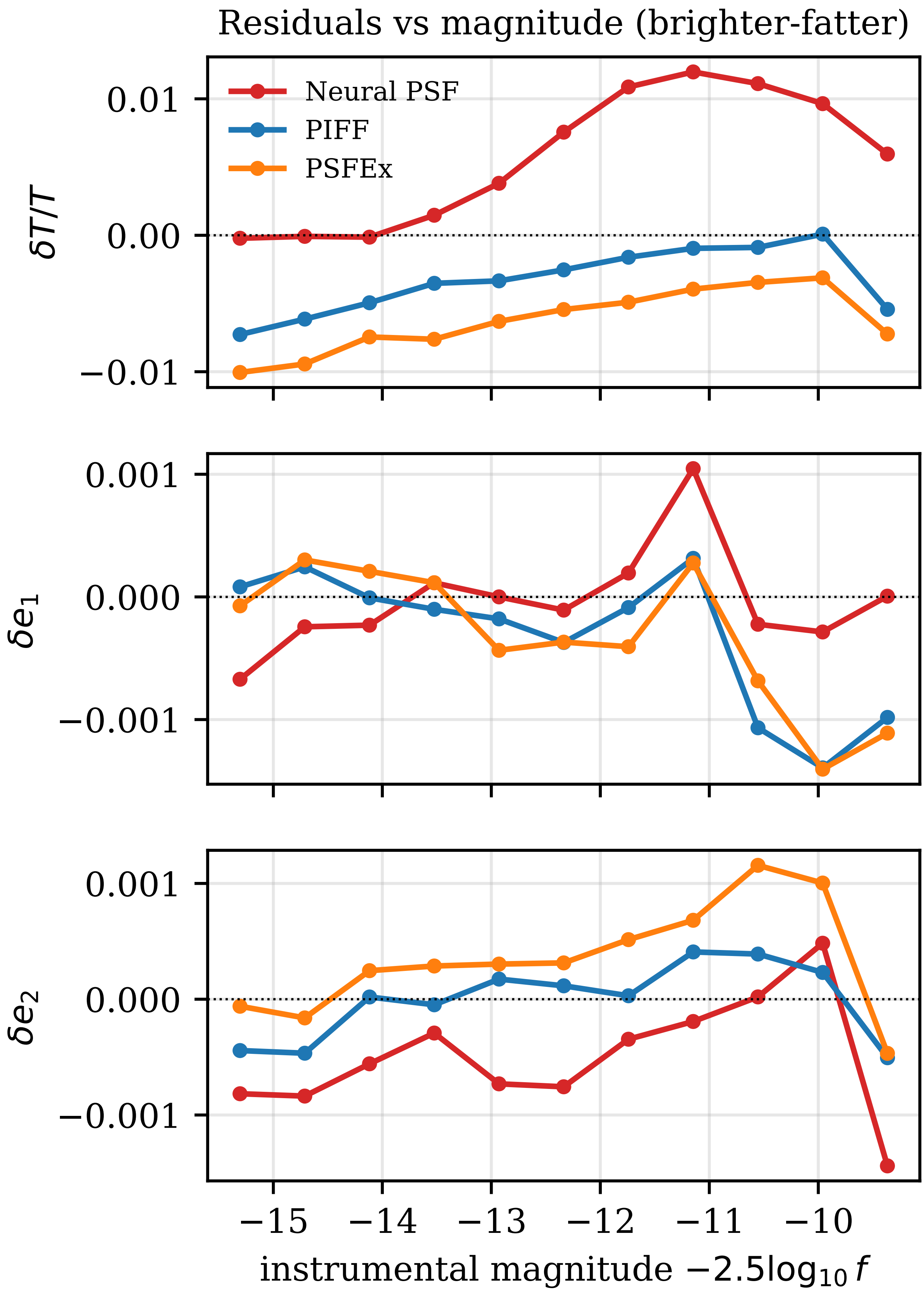

Figure 14. Residuals versus instrumental magnitude (cf. PIFF Fig. 10). Our model (red) carries a magnitude-dependent size residual peaking at +1.2% (§5).

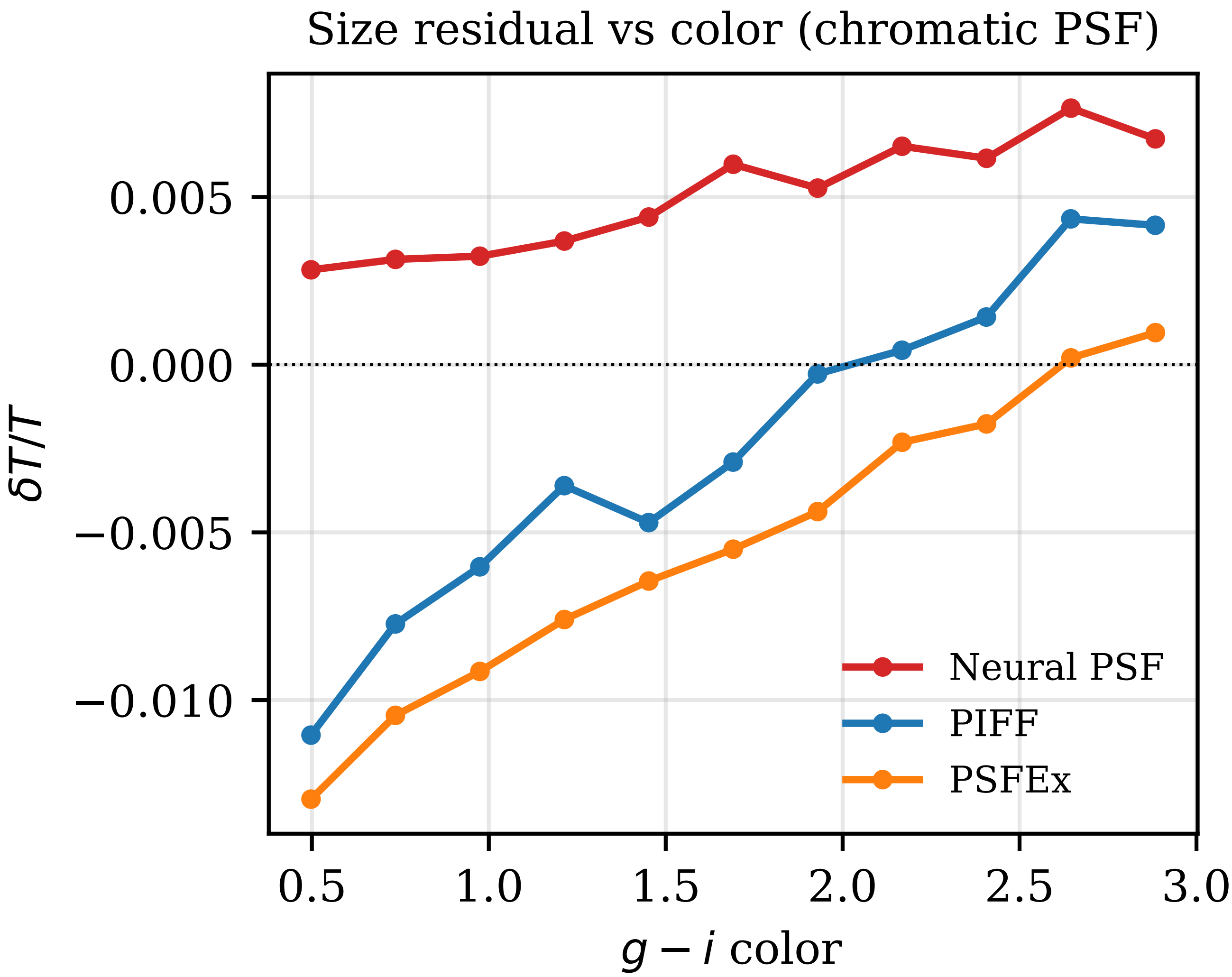

Figure 15. Size residual versus $g - i$ color (cf. PIFF Fig. 11); color conditioning keeps our chromatic residual flat.

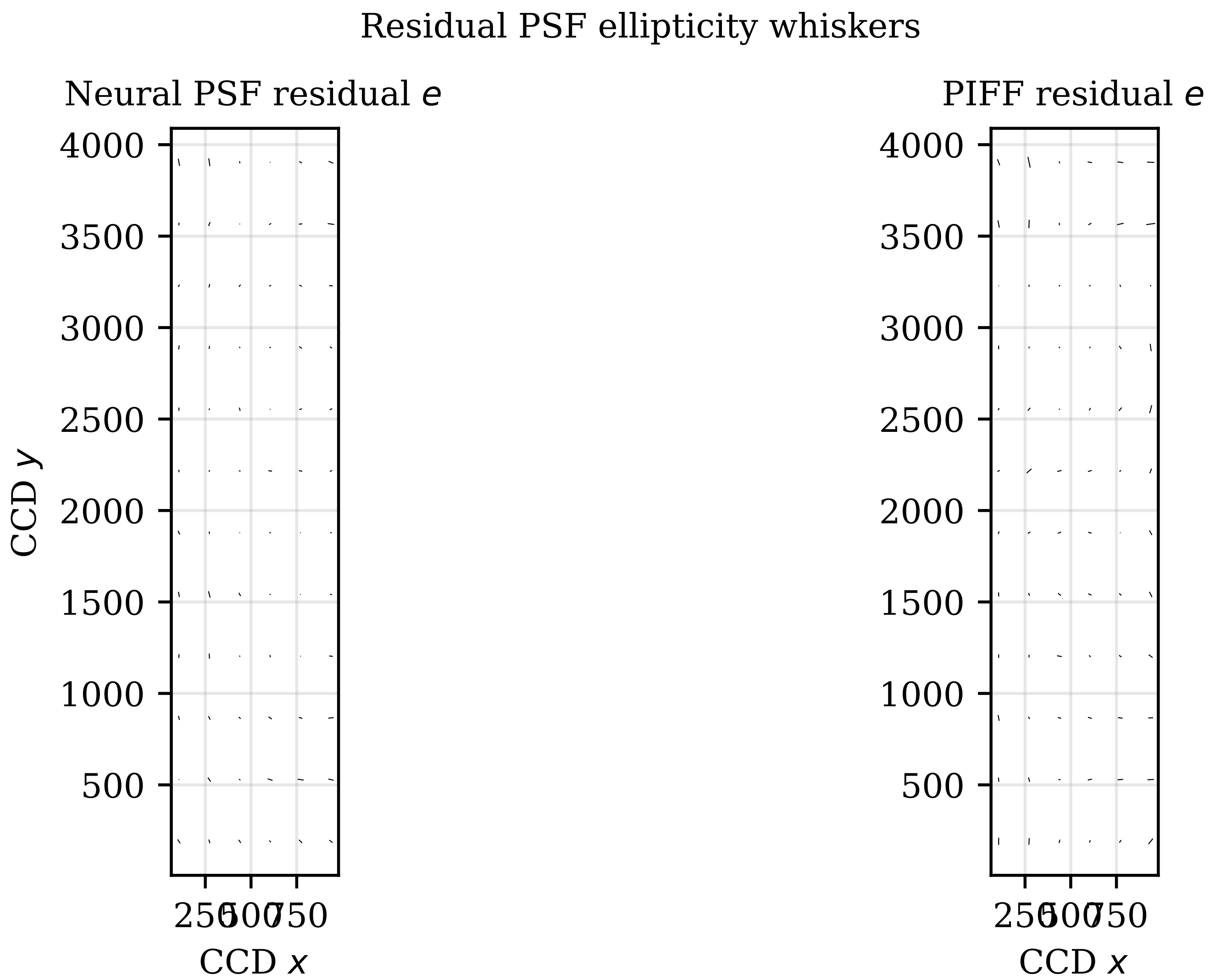

Figure 16. Residual PSF-ellipticity whiskers across the CCD, Neural PSF and PIFF (cf. PIFF Fig. 2).

REFERENCES

Anderson, J., & King, I. R. 2000, PASP, 112, 1360, doi: 10.1086/316632

Astropy Collaboration. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74

Barbary, K. 2016, JOSS, 1, 58, doi: 10.21105/joss.00058

Bernstein, G. M., & Jarvis, M. 2002, AJ, 123, 583, doi: 10.1086/338085

Bertin, E. 2011, ADASS XX, 442, 435

Booth, J. G., & Hobert, J. P. 1999, Journal of the Royal Statistical Society: Series B, 61, 265

Delyon, B., Lavielle, M., & Moulines, E. 1999, Annals of Statistics, 27, 94

Dempster, A. P., Laird, N. M., & Rubin, D. B. 1977, Journal of the Royal Statistical Society: Series B, 39, 1

Jarvis, M., et al. 2016, MNRAS, 460, 2245

Jarvis, M., et al. 2021, MNRAS, 501, 1282, doi: 10.1093/mnras/staa3679

Lauer, T. R. 1999, PASP, 111, 227, doi: 10.1086/316319

Levine, R. A., & Casella, G. 2001, Journal of Computational and Graphical Statistics, 10, 422

Liaudat, T. I., Starck, J.-L., & Kilbinger, M. 2023, Frontiers in Astronomy and Space Sciences, 10, 1158213, doi: 10.3389/fspas.2023.1158213

Paszke, A., et al. 2019, in NeurIPS

Rowe, B. 2010, MNRAS, 404, 350

Rowe, B. T. P., et al. 2015, Astronomy and Computing, 10, 121, doi: 10.1016/j.ascom.2015.02.002

Vaswani, A., Shazeer, N., Parmar, N., et al. 2017, in Advances in Neural Information Processing Systems (NeurIPS), Vol. 30

Wei, G. C. G., & Tanner, M. A. 1990, Journal of the American Statistical Association, 85, 699